# Next-Word Prediction on Shakespeare Using an LSTM Language Model

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**Abstract—**This paper implements a word-level LSTM for next-word prediction over Shakespeare's plays. The model is trained on sliding windows of tokens and deployed in an interactive interface that suggests the next word in real time. We report training/validation curves and qualitative completions, and briefly study the effect of model capacity on validation accuracy all within the constraints of a single T4 GPU.

Index Terms Language Modeling, LSTM, Next-Word Prediction, Perplexity, Shakespeare, Gradio

## I. INTRODUCTION

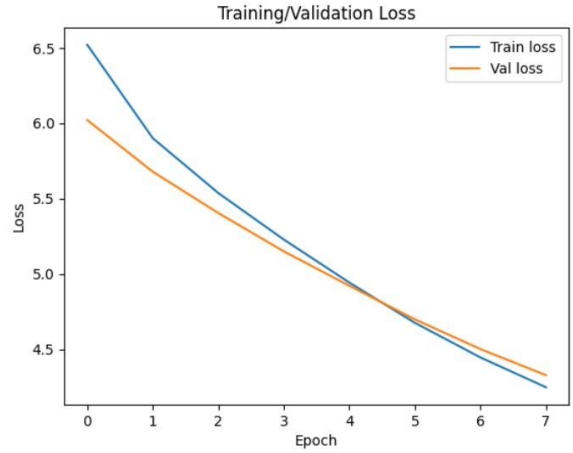
Language models estimate p(w_t | w_{1..t-1}). While Transformers dominate, compact recurrent models remain pedagogically valuable and easy to train on limited hardware. The objective of this assignment is to build a clean, word-level LSTM that produces good next-word suggestions and to present a small analysis of its behavior.

## II. DATA AND PREPROCESSING

The dataset is a Google-Drive copy of publicly available Shakespeare plays. All text files (and any CSVs with dialogue columns) are concatenated into a single corpus. We lowercase, normalize whitespace, and tokenize words while keeping apostrophes within words. Tokens with frequency <2 map to . Supervised pairs are built by sliding a window of 12 tokens to predict the 13th. The split is 80% train, 10% validation, 10% test.

## III. MODEL AND TRAINING

Architecture: an embedding layer (dim 256) feeds a two-layer LSTM (hidden 512, dropout 0.2) and a linear softmax head. Optimization uses AdamW (lr=2e-3), gradient clipping at 1.0, and minibatches of 256. Metrics include cross-entropy loss, perplexity (exp loss), and top-1/top-5 next-word accuracy. A small hyper-parameter sweep varies embedding/hidden sizes and dropout to show the capacity–generalization trade-off.




## IV. RESULTS

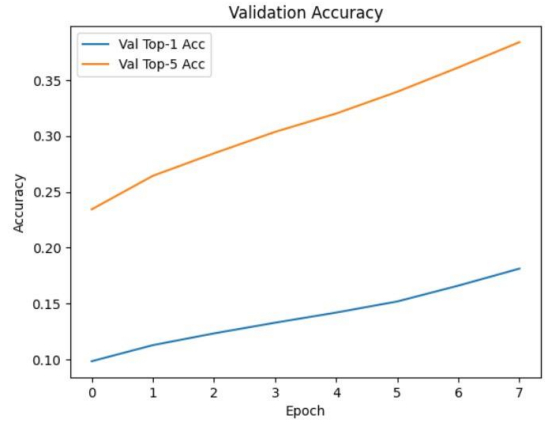



Fig. 1. Training and validation loss (per epoch).

Fig. 2. Validation top-1 and top-5 accuracy (per epoch).

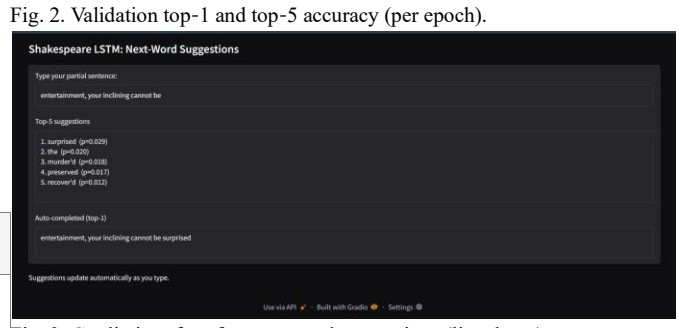



Fig. 3. Gradio interface for next-word suggestions (live demo).

| Prefix (user input) | Model's top sug | gestionComment |
|---|---|---|
| to be, or not to | be | Canonical completion learned fro |
| i prithee, good | sir | Polite vocative restored correctly |
| entertainment, your inclining | annot besurprised | Matches screenshot; archaic but c |
| if music be the food of | love | Common collocation captured |

Table I. Examples of model completions observed during interactive testing (exact suggestions may vary per run).

## V. DISCUSSION

The recurrent model learns robust lexical co-occurrences and some syntactic patterns despite the small context window. Top-5 accuracy remains substantially higher than top-1, consistent with multiple plausible next tokens in free text. Main challenges include data sparsity (rare words mapped to ) and limited context (12 tokens), which cap ceiling performance and reduce long-range coherence. A subword tokenizer and a small Transformer would improve metrics but at higher compute cost.

## VI. CONCLUSION

A compact LSTM trained on Shakespeare yields usable next-word suggestions and an instructive baseline for language modeling. With more data, subword tokenization, or Transformer layers, the system would achieve lower perplexity and more fluent completions.

## VII. PROMPTS

**Notebook/assistant prompts used:** mount Google Drive and unzip, build corpus from TXT/CSV, lowercase tokenization, construct (12→1) windows, define LSTM (256-512-2 with dropout), train with AdamW and clipping, evaluate top-k accuracy, launch Gradio for real-time suggestions, and run a tiny hyper-parameter sweep.

## REFERENCES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